MIL-T-43259C
29 September 1978
SUPERSEDING
MIL-T-43259B
10 February 1971

MILITARY SPECIFICATION

TRAYS, SERVICE, PLASTIC (LAMINATED FIBER GLASS REINFORCED POLYESTER RESIN)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and two sizes of serving tray made of laminated fiber glass and polyester resins.

1.2 Classification. The trays shall be of the following sizes, as specified (see 6.2).

Size 1 - 14-1/8 x 18 x 7/8 inch deep (359 x 457 x 22 mm)
Size 2 - 15-1/4 x 20-1/2 x 7/8 inch deep (387 x 521 x 22 mm)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-P-553 - Paper, Wrapping, Tissue.
PPP-B-601 - Box, Wood, Cleated-Plywood.
PPP-B-621 - Box, Wood, Nailed and Locked-Corner.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7350

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PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

DRAWINGS

US ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

8-2-223 - Tray Service Plastic; Laminated Fiberglass, Reinforced Polyester Resin, Size 1.
8-2-224 - Tray Service Plastic; Laminated Fiberglass, Reinforced Polyester Resin, Size 2.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The trays delivered under this specification shall be the manufacturer's current commercial standard product, except for any changes necessary to comply with specification requirements. Standard color control samples shall be available (see 6.3).

3.1.1 Standard sample. The dyed tape and webbing shall match the standard sample for shade and shall be equal to, or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Materials. Resin-bonded type E fiberglass mats and veils shall be used in the fabrication of trays (see 3.4). All material shall be free from dirt and metal particles. Laminating resins shall be of the unsaturated polyester type, thermosetting, containing a monomeric styrene as a solvent (see 6.5) and cured with an appropriate catalyst (see 6.7).

3.3 Design and dimensions. The tray shall conform to the design and dimensions shown on Drawing 8-2-223, or Drawing 8-2-224, as applicable. Stacking lugs shall be molded as an integral part of the tray, on the outside of the sides, not on the rim. Stacking lugs shall be rectangular and not less than 1/4 inch by 3/8 inch (6.4 by 9.5 mm) but not more than 0.040 inch (1 mm) in thickness. Eight stacking lugs shall be used, two at each corner.

3.4 Construction. The tray shall be of laminated construction consisting of a layer of uncolored continuous filament strand glass fiber surfacing mat, three or more layers of 1-1/2 ounce (43 gm) glass fiber chopped strand reinforcing mat, or a glass fiber preform of equal weight, and one layer of uncolored continuous filament strand glass fiber surfacing mat. The layers shall be bonded together, in the order specified, under heat and pressure, using the polyester laminating resin specified in 3.2. Trays shall not be made with a gel resin overlay or any thick resin surface or surfacing material. Alternately, trays may be made by replacing the colored glass fiber veil with a continuous filament strand glass fiber bonded surfacing mat or veil which has been printed to give a pattern resembling the standard color control samples (see 6.3). The color and pattern shall meet the requirements of 3.11.

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3.5 Resistance to impact. The trays shall show no rupture, separated pieces, or open cracks when tested as specified in 4.3.1.

3.6 Resistance to boiling water. The tray shall show no cracks, changes of color, or blooming of fibers and shall not warp diagonally more than 0.093 inches (2.4 mm) when tested as specified in 4.3.3.

3.7 Cure test. The tray shall show a minimum average dial reading of 30 when tested as specified in 4.3.2 with the Barcol Impressor.

3.8 Warpage. Warpage shall be not greater than 0.0625 inches (1.6 mm) when tested as specified in 4.3.4 and 4.3.4.1.

3.9 Resistance to dry heat. The tray shall show no cracks or blisters when tested as specified in 4.3.5.

3.10 Odor. The tray shall have no objectionable odor as received (see 4.3.6.1) and shall develop no objectionable odor, after testing with boiled distilled water as specified in 4.3.6.2.

3.11 Color, color pattern and shade. Unless otherwise specified (see 6.2), the color of the trays viewed from approximately 5 feet (1.5 m), shall be tan and the color pattern and shade shall be within the range of variations of the standard color control samples No. 666 and No. 689 (see 6.3).

3.12 Finish. The surfaces of trays shall have a uniform high luster equal to or higher than the standard color control samples (see 6.3) produced by molding techniques and not by application of any finish or protective coating. After molding, the edges shall be smooth and free from sharpness or irregularities. These edges may be treated by sanding or burnishing.

3.13 Marking for identification. Each tray shall be permanently marked with the manufacturer's name or trademark, in accordance with the manufacturer's commercial practice, on the outside bottom surface of the tray.

3.14 Workmanship. The finished tray shall have smoothly rounded edges, shall be uniform in texture, finish, color, and shade (within specified tolerance limits). The tray shall be clean and free from cracks, breaks, abraded areas, areas of delamination, scratches, blisters, flow marks, chips, dents and unfilled areas and foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except where otherwise indicated hereinafter, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Examination of end item. The lot shall be the number of trays of the same size submitted for examination at one time and the sample unit for the examination of the end item shall be one tray.

4.2.2.1 Visual examination. The finished tray shall be examined for defects in appearance, construction and general workmanship as specified in table I. The inspection level shall be level I with an AQL of 6.0 defects, expressed in terms of defects per hundred units.

TABLE I. Visual defects

Examine	Defect
Appearance	Scuffed or abraded area. Scratch. Dent. Chalking. Pinhole. Blister. Wrinkle. Pit. Flow mark.

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TABLE I. Visual defects (cont'd)

Examine	Defect
Appearance (cont'd)	Unfilled area. Chipped. Presence of mechanical oil or grease, dirt, wood shaving, metal chips, or other foreign matter on the tray surface.
Luster	Lacks luster (dull). Not uniform high luster throughout. Not equal to or higher than that on standard color samples (see 6.3).
Color	Not specified color. Lack of uniformity of color and shade within established tolerances. Shade of color does not match the standard samples (see 6.3).
Texture and finish	Presence of any finishing coating. Lack of uniformity in texture and finish. NOTE: Texture may be considered as referring to the fine detail of structure.
Construction and workmanship	Varies from design and shape shown on applicable drawing. Any area of delamination.
Material	Presence of molded-in particles of metal, cured resin flash, dirt, or other materials.
Corners and edges	Sharp. Flash not "broken" and polished.
Surfaces and curves	Not blended smoothly.
Stacking feature	A unit does not stack firmly and evenly with another unit. Poor nesting resulting in restricted air circulation (for drying purposes).

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TABLE I. Visual defects (cont'd)

Examine	Defect
Identification marking	Not permanently marked as specified. Missing. Illegible. Not on outside bottom surface.

4.2.2.2 Dimensional examination. Any dimension that is not within the specified limits shall be a defect. The inspection level shall be level S-2 with an AQL of 10.0 defects, expressed in terms of defects per hundred units.

4.2.2.3 Preservation inspection. An examination shall be made to determine that applicable preservation, packing and marking comply with the section 5 requirements. Defects shall be in accordance with table II. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 2.5 defects per hundred units.

TABLE II. Examination of packaging

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, defective or not as specified.
Workmanship	Inadequate application of components. Bulging or distorted container.
Content	Number per interior pack and per container is more or less than required.

4.2.3 Testing of the end item. The end item shall be tested for the characteristics listed in table III. The sample unit shall be six finished

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trays. The inspection level shall be S-2. The AQL for each characteristic shall be 4.0 defects per hundred units. All test reports shall contain the individual values utilized in expressing the final result.

TABLE III. Instructions for testing

Characteristic	Requirement paragraph	Test method	Number of determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Resistance to impact	3.5	4.3.1	4	X	
Resistance to boiling water	3.6	4.3.3	1	X	
Cure test	3.7	4.3.2	Average of 10		Dial reading
Warpage	3.8	4.3.4 and 4.3.4.1	1	X	
Resistance to dry heat	3.9	4.3.5	1	X	
Odor	3.10	4.3.6	1	X	

4.3 Test methods.

4.3.1 Impact resistance test. The tray in the position of normal use shall be supported at each of its four corners at a point measured in three inches from each edge for each size of tray. The supports shall be wooden blocks 1 inch square and 3 inches high (25 by 76 mm). The upper ends of the blocks shall be rounded to approximately 1/2 inch (13 mm) radius. An iron or steel ball weighing 8 pounds (3.6 kg) (see 6.6) shall be dropped onto the center of the tray. If the ball bounces it must be caught by hand so that it strikes the tray only once. This drop shall be repeated three times giving a total of four drops. The tray shall be examined after each drop for any signs of rupture, separated pieces, or open cracks. The height of the drop shall be 16 inches (406 mm) for size 1 trays and 18 inches (457 mm) for size 2 trays. The height shall be measured from the bottom of the ball to the point of impact of the ball against the tray. Both surfaces of the tray shall be examined for

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rupture, separated pieces, or open cracks after each impact as damage may occur at the surface opposite to the side impacted. A failure of any test determination to meet the requirement shall be scored a defect for the test sample unit.

4.3.2 Cure test. Measurements for degree of cure shall be made using the Barcol Impressor. Five readings shall be taken at each of the four corners and at the center of the tray on both the inside and bottom surfaces. A metal block placed beneath the surface being tested will give the necessary support needed for accurate readings. The average of five readings of each test point shall be not less than the requirements.

4.3.3 Resistance to boiling water. Each sample tray shall be immersed by suspending for 30 minutes in boiling water, removed, and allowed to stand for 1 hour at room temperature. This cycle shall be immediately repeated three times for a total of four such cycles. The sample shall then be allowed to stand overnight in air at normal room temperature before examination for cracks, changes in color, or blooming of fibers when compared with an unexposed tray from the same lot. Diagonal warpage shall be determined by placing the tray upright (position of normal use on the table) on a flat surface, either raised corner shall be held down by hand so that three corners are touching. In this position, the clearance between the raised corner of the tray and the flat surface shall be less than the thickness of a 0.093 inch (2.4 mm) feeler gage.

4.3.4 Warpage. Rim warpage shall be determined by placing the tray upside down on a flat surface and holding it firmly in place by a 10 pound (4.6 kg) weight with a diameter not exceeding 5 inches (127 mm) positioned at the center of the tray. In this position the tray shall be sufficiently planed so that the clearance between the tray and the flat surface is less than the thickness of a 0.0625 inch (1.6 mm) feeler gage.

4.3.4.1 Warpage (other than rim). Trays shall be placed upright (position of normal use on the table) on a flat surface and loaded with a 10 pound (4.6 kg) weight as specified in 4.3.4. In this position the tray shall be sufficiently planed so that the clearance between the tray and the flat surface is less than the thickness of a 0.0625 inch (1.6 mm) feeler gage.

4.3.5 Resistance to dry heat. Place the tray in a circulating air oven at a temperature between 170 and 173 F (77 and 78 C) for 8 hours. The tray shall then be removed, placed in the position of normal use on a wooden or other non-conducting table and allowed to cool normally to room temperature before examination for cracks or blisters.

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4.3.6 Odor.

4.3.6.1 As received. The tray shall be unpacked, unstacked, and left exposed at room temperature for not less than 5 minutes. Any objectionable odor or a strong styrene odor after the 5 minute exposure period shall constitute a failure.

4.3.6.2 After testing. The tray shall be half filled with boiling distilled water. A sheet of plate glass shall be immediately placed over the rim of the tray and weighted down to obtain good contact between the rim and the glass. After 5 minutes the glass shall be removed and the presence or absence of odor shall be determined after waiting 60 seconds.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Twelve trays of one size only, shall be nested with a sheet of tissue paper laid between and overlapping the outer edges of each tray. The tissue paper shall conform to type I, class 1 or 2 of UU-P-553. The stack of twelve nested trays shall then be inserted into a fiberboard box conforming to style OSC, type CF, variety SW, class weather-resistant, grade W5c of PPP-B-636, except that the outer flaps shall overlap a minimum of 1-1/2 inches (38 mm). The inside dimensions of each box for size 1 trays shall be 14-1/4 inches in length, 4 inches in width and 18-1/4 inches in depth (362 by 102 by 464 mm) and for size 2 trays shall be 15-1/2 inches in length, 4 inches in width and 20-3/4 inches in depth (394 by 102 by 527 mm). Box closure shall be effected along the entire length of the flaps by sealing with 2-inch (51 mm) minimum width gummed paper tape conforming to type III, grade B of PPP-T-45, or as specified in the appendix of PPP-B-636.

5.1.2 Commercial. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

* 5.2.1 Level A packing. Thirty-six trays of one size only, preserved as specified in 5.1, shall be packed in a shipping container conforming to PPP-B-601, overseas type, style optional or PPP-B-621, class 2, style 4. Level A unit packs shall be packed flat, three in depth within a shipping container. Container closure and reinforcement shall be as specified in the appendix of applicable specification.

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* 5.2.2 Level B packing. Thirty-six trays of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class weather-resistant, grade V3c, V3s or V4s of PPP-B-636. Level A unit packs shall be packed flat, three in depth within a shipping container. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 and MIL-STD-1188, as applicable.

6. NOTES

6.1 Intended use. The trays covered by this specification are intended for use in cafeterias, mess halls, and hospitals with food contained in serving dishes and not in direct contact with the tray.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Color required if other than specified (see 3.11).
(d) Selection of the applicable degree of preservation and packing (see 5.1 and 5.2).

6.3 Color control. A set of standard color control samples will be available at the procuring agency for examination by bidders. These will be used for both regular construction and alternate construction trays. Standard samples No. 666 and No. 689 should be obtained by the procuring agency from the US Army Natick Research and Development Command, Natick, MA 01760.

6.4 For information only. Acceptable trays have been molded using the following commercially available resin solutions, namely, 90 parts by weight of a semi-rigid polyester resin and 10 parts by weight of a flexible polyester resin. This blend contains 29 to 31 percent of monomeric styrene and 69 to 71 percent of polyester resin. The trays were cured with 1 percent of bensoyl peroxide (see 3.2 and 3.4). Other resins or combination of resins and other catalysts may be used provided they product trays meeting the requirements of this specification.

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6.5 Odor. The proportion of styrene used in producing the tray should be kept as low as possible in order to minimize the styrene odor. This odor will disappear after the first or second washing.

6.6 Weights. The 8 pound (3.6 kg) iron or steel ball specified in 4.4.1.1 can be readily obtained from distributors of athletic goods. (Standard balls for "putting-the-shot" weigh 8 and 12 pounds (3.6 and 5.4 kg).

* 6.7 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

* 6.8 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practice, conversion factors and symbols specified in ASTM E 380 Standard for Metric Practice and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.9 Environmental. Environmental pollution prevention measures are contained in the packaging material specification referenced herein. Refer to material specification or preparing activity for recommended disposability methods.

6.10 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-T-43259C Trays, Service, Plastic (Laminated Fiber Glass Reinforced Polyester Resin)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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